MSK Histology — NBME Study Guide

Muscle · Cell junctions & cytoskeleton · Bone & cartilage · Connective tissue & ECM · Skin · Slide identification

Red = extra high yield (classic NBME) Black = core content *Italic blue = mnemonic*

1. Muscle

	Skeletal	Cardiac	Smooth
Nuclei	**Many, peripheral**	**1–2, central**	1, central (cigar/corkscrew when contracted)
Striations	Yes	Yes	No
Special	Triads (T-tubule at **A–I junction** + 2 terminal cisternae)	**Intercalated discs (fascia adherens, desmosomes, gap junctions); dyads at Z-line; branched**	Dense bodies (α-actinin), caveolae, gap junctions
Regeneration	Satellite cells (limited)	**None → fibrous scar**	Good (mitosis, pericytes)
Ca^{2+} binds	Troponin C	Troponin C	**Calmodulin → MLCK**

- Organization: epimysium (whole muscle) → perimysium (fascicles) → **endomysium** (each fiber; reticular fibers) → fiber → myofibrils → sarcomeres.
- **Sarcomere = Z to Z line. A band (myosin, entire length) stays constant. I band (actin only), H zone (myosin only) shorten during contraction. M line = middle of H zone.** *(HIZ shrinkage; A is Always the same)*
- Thin filament = actin + tropomyosin + troponin (T binds tropomyosin, I inhibits, C binds Ca^{2+}). Thick = myosin II.
- **Accessory proteins**: **titin** (Z to M; elasticity, centers myosin; largest protein), **nebulin** (molecular ruler for actin length), **α-actinin** (anchors actin at Z), **desmin** (links Z discs to each other and sarcolemma), **dystrophin** (actin → dystroglycan → laminin; Duchenne/Becker), myomesin (M line), CapZ (+ end), tropomodulin (− end).
- Fiber types: **type I = slow oxidative, red (myoglobin, mitochondria), posture, marathon** *("one slow red ox")*; type IIa fast oxidative-glycolytic; **type IIb/x = fast glycolytic, white, sprint**.
- Muscle spindle (intrafusal: nuclear bag/chain; γ motor; senses length) vs Golgi tendon organ (tendon; senses tension; Ib).
- **Satellite cells**: muscle stem cells under the basal lamina; hypertrophy (not hyperplasia) after exercise.

2. Cell junctions & cytoskeleton

Junction	Proteins	Cytoskeleton	Function / disease
Tight (zonula occludens)	**Claudins, occludin**	**Actin**	**Seal paracellular space; apical/basolateral polarity; C. perfringens toxin targets claudins**
Adherens (zonula adherens)	**E-cadherin (Ca^{2+}-dependent), catenins**	**Actin**	**Belt; loss of E-cadherin → metastasis, diffuse gastric/lobular breast cancer**
Desmosome (macula adherens)	**Desmogleins, desmocollins (cadherins), plakoglobin, desmoplakin**	**Intermediate (keratin)**	**Spot weld; pemphigus vulgaris = anti-desmoglein 1/3 → intraepidermal, flaccid bullae, + Nikolsky**
Gap junction	Connexins (connexon)	—	Electrical/metabolic coupling (cardiac, smooth muscle); Charcot–Marie–Tooth X-linked (connexin 32)
Hemidesmosome	**Integrins (α6β4), BP180/BP230**	**Intermediate**	**Cell → basement membrane; bullous pemphigoid = anti-hemidesmosome → subepidermal tense bullae, Nikolsky −**
Focal adhesion	Integrins → fibronectin	Actin	Cell–ECM; migration signaling

- **Intermediate filaments = tumor markers: cytokeratin (epithelium/carcinoma), vimentin (mesenchyme/sarcoma), desmin (muscle), GFAP (glia), neurofilament (neurons), lamins (nucleus).**

- Microvilli = actin core (absorptive surface); stereocilia = long actin microvilli (epididymis, inner ear hair cells); cilia = microtubules 9+2 with dynein (Kartagener).
- Basement membrane = basal lamina (type IV collagen, laminin, heparan sulfate, nidogen) + reticular lamina (type III).

3. Cartilage

Type	Collagen	Perichondrium	Where
Hyaline	**Type II**	**Yes (except articular)**	**Articular surfaces, costal, trachea, larynx (most), nose, fetal skeleton & epiphyseal plate**
Elastic	**Type II + elastic fibers**	**Yes**	**External ear, epiglottis, auditory tube**
Fibrocartilage	**Type I + II**	**No**	**Intervertebral disc (annulus), pubic symphysis, menisci, TMJ, tendon insertions**

- **Cartilage is avascular** (nutrients diffuse from perichondrium/synovial fluid) → poor healing.
- Chondrocytes in lacunae; **isogenous groups** (from one cell). Matrix: aggrecan (chondroitin + keratan sulfate on hyaluronan) → water binding → compression resistance.
- **Growth: appositional (from perichondrium chondroblasts) vs interstitial (chondrocytes dividing inside; growth plate).**
- Osteoarthritis = articular cartilage degeneration (no perichondrium to repair).

4. Bone

- Matrix: **type I collagen** (osteoid) + hydroxyapatite. Cells:
  - **Osteoprogenitor** (mesenchymal, RUNX2) → **osteoblasts** (cuboidal, on surfaces; make osteoid, **alkaline phosphatase**, osteocalcin; express **RANKL** and OPG; PTH receptors).
  - **Osteocytes** (trapped in lacunae, connected by canaliculi via gap junctions; mechanosensors; sclerostin).
  - **Osteoclasts**: multinucleated, from **monocyte/macrophage lineage**; ruffled border + sealing zone (integrins) form **Howship lacuna**; secrete H^+ (carbonic anhydrase II) and cathepsin K; **tartrate-resistant acid phosphatase (TRAP)**. Activated by **RANKL–RANK** + M-CSF; OPG = decoy receptor; denosumab = anti-RANKL. Calcitonin inhibits.
- **PTH acts on osteoblasts (receptors) → ↑ RANKL → osteoclast activation.** Intermittent PTH (teriparatide) builds bone.
- **Compact (cortical) bone**: osteons/Haversian systems — concentric lamellae around a Haversian canal; **Volkmann canals** run perpendicular, connecting them; interstitial and circumferential lamellae. **Spongy (cancellous)**: trabeculae, red marrow, no osteons.
- Woven (primary, immature; random collagen; fracture callus, Paget) vs lamellar (mature).
- Periosteum (outer fibrous + inner osteogenic layer; Sharpey fibers) vs endosteum.
- **Ossification: intramembranous (flat skull bones, clavicle, mandible — mesenchyme → bone directly) vs endochondral (long bones, vertebrae, base of skull — hyaline cartilage model replaced).**
- **Growth plate zones (epiphysis → diaphysis): reserve (resting) → proliferation (stacked coins; achondroplasia FGFR3) → hypertrophy (chondrocytes swell, Ca^{2+} matrix; vitamin D deficiency/rickets here) → calcification → ossification.**
- Fracture repair: hematoma → soft callus (fibrocartilage) → hard callus (woven bone) → remodeling (lamellar).

Disorder	Defect	Labs / clue
Osteoporosis	**↓ bone mass, normal mineralization**	**Normal Ca, PO_4, ALP, PTH; vertebral crush, hip, Colles fx**
Osteopetrosis	**Osteoclast failure (carbonic anhydrase II)**	**Dense brittle bone, marrow failure, Erlenmeyer flask**
Osteomalacia / rickets	**Defective mineralization (vit D)**	**↓ Ca, ↓ PO_4, ↑ ALP, ↑ PTH; rachitic rosary, bowed legs**
Paget disease	**Excess osteoclast then blast activity**	**Isolated ↑↑ ALP, mosaic woven bone, ↑ hat size, hearing loss**

Osteogenesis imperfecta	Type I collagen	Blue sclerae, multiple fractures, hearing loss
Achondroplasia	FGFR3 gain (AD) — ↓ proliferation zone	Short limbs, normal trunk; ↑ paternal age

5. Connective tissue & ECM

- Cells: fibroblasts (main), macrophages, mast cells (heparin, histamine; IgE), plasma cells, adipocytes. Fibers: collagen, reticular (type III), elastic. Ground substance: GAGs, proteoglycans, glycoproteins (fibronectin, laminin).
- Loose (areolar, under epithelia) vs dense irregular (dermis, capsules) vs **dense regular (tendons, ligaments — parallel type I, few fibroblasts, poor blood supply)**.

Collagen	Location	Disease
I (most abundant)	**Bone, skin, tendon, dentin, fascia, cornea, late wound repair**	**Osteogenesis imperfecta, Ehlers–Danlos (some)**
II	**Cartilage, vitreous, nucleus pulposus**	**Achondrogenesis/Stickler**
III (reticulin)	**Skin, vessels, uterus, granulation tissue, lymphoid organs**	**Vascular Ehlers–Danlos**
IV	**Basement membrane**	**Alport (hearing, eye, kidney), Goodpasture (anti-α3 IV)**

"Be (So Totally) Cool, Read Books": type I Bone; II Cartilage; III Reticulin; IV Basement membrane.

- Elastin: rich in glycine, proline, **desmosine** cross-links (lysyl oxidase), scaffold = **fibrillin-1**. **Marfan = fibrillin-1 (FBN1)**: tall, arachnodactyly, lens subluxation upward, aortic root dilation/dissection, MVP. α_1-antitrypsin deficiency → elastase destroys lung elastin (panacinar emphysema).
- Collagen synthesis & disease details → MSK Biochemistry guide.

6. Skin

- **Epidermis layers (deep → superficial): stratum basale (stem cells, melanocytes) → spinosum (desmosomes "spines", Langerhans cells) → granulosum (keratohyalin, lamellar bodies) → lucidum (thick skin only) → corneum (dead keratinized)** *("Californians Like Girls in String Bikinis": corneum, lucidum, granulosum, spinosum, basale)*.
- Cells: keratinocytes (~90%), **melanocytes (neural crest; melanin to keratinocytes via melanosomes; albinism = normal number, ↓ tyrosinase; vitiligo = ↓ number)**, **Langerhans cells (antigen-presenting; Birbeck "tennis racket" granules; CD1a, S-100; bone marrow origin)**, **Merkel cells** (touch, basale).
- Dermis: papillary (loose CT, Meissner corpuscles) and reticular (dense irregular, Pacinian corpuscles deeper, glands). Hypodermis = adipose.
- Thick skin (palms/soles): stratum lucidum, no hair. Thin skin: hair follicles, sebaceous glands.
- Glands: **eccrine sweat (merocrine; everywhere; sympathetic cholinergic)**, apocrine (axilla, groin; open into follicles; after puberty), sebaceous (holocrine; acne).
- Receptors: Meissner (light touch, papillary dermis, glabrous), Pacinian (vibration/pressure, deep, onion layers), Merkel (sustained pressure), Ruffini (stretch), free nerve endings (pain, temperature).
- Psoriasis: thickened spinosum (acanthosis), parakeratosis, Munro microabscesses. Burns: 1st degree epidermis; 2nd into dermis (blisters); 3rd full thickness (painless).

7. Slide identification tips

- **Peripheral multiple nuclei + striations = skeletal; central nucleus + branching + intercalated discs = cardiac; spindle cells with central nucleus, no striations = smooth.**
- **Concentric lamellae around a central canal = osteon (compact bone). Lacunae with isogenous groups = cartilage (glassy = hyaline; dark fibers = elastic; rows of cells between collagen = fibrocartilage).**
- Large multinucleated cell in a Howship lacuna = osteoclast; cuboidal row on osteoid = osteoblasts.
- Parallel collagen with flattened nuclei in rows = tendon (dense regular).

- Keratinized stratified squamous with thick corneum and lucidum = thick skin.